2110201: COMP ENG MATH (2023/2) Quiz 7

Name: TA. Khun Student ID: Entry Number:

1. (3 points) Find **all** eigenvalues along with their corresponding eigenvectors of $\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$.

Key: $|A-\lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & -2 & 2 \\ -2 & 1-\lambda & -2 \\ 2 & -2 & 1-\lambda \end{vmatrix} = 0$$

$$[(1-\lambda)^3+16]-3[4(1-\lambda)] = 0$$

$$1-3\lambda+3\lambda^2-\lambda^3+16-12+12\lambda = 0$$

$$-\lambda^3+3\lambda^2+9\lambda+5 = 0$$

$$\lambda^3-3\lambda^2-9\lambda-5 = 0$$

$$\begin{array}{r|rrrr} -1 & 1 & -3 & -9 & -5 \\ & & -1 & 4 & 5 \\ \hline & 1 & -4 & -5 & 0 \end{array}$$

$\therefore (\lambda+1)(\lambda^2-4\lambda-5)=0$

$(\lambda+1)^2(\lambda-5)=0$

$\therefore \lambda=-1,5$

$\lambda=-1$

$$\begin{bmatrix} 2 & -2 & 2 \\ -2 & 2 & -2 \\ 2 & -2 & 2 \end{bmatrix} \xrightarrow{RRE} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad E_{-1} = \text{span}\left(\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\right\}\right)$$

$\lambda=5$:

$$\begin{bmatrix} -4 & -2 & 2 \\ -2 & -4 & -2 \\ 2 & -2 & -4 \end{bmatrix} \xrightarrow{RRE} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad E_5 = \text{span}\left(\left\{\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}\right\}\right)$$

$\therefore$ Ans: $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}/-1$, $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}/-1$, and $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}/5$

2. (6 points, 2 points each) Given an SVD of a matrix $\mathbf{A}$ as following:

$$\underbrace{\begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix}}_{\text{col}} \begin{bmatrix} \sqrt{30} & 0 & 0 \\ 0 & 2\sqrt{5} & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} & 0 & -2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}^{\mathbf{T}}$$

row nul.

2.1. Find the **basis** of the following subspaces:

a. **row(A)**

$$\left\{\begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}\right\}$$

b. **col(A)**

$$\left\{\begin{bmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix}, \begin{bmatrix} 1/\sqrt{10} \\ -3/\sqrt{10} \end{bmatrix}\right\}$$

c. **nul(A)**

$$\left\{\begin{bmatrix} -2/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}\right\}$$

d. **nul($\mathbf{A^T}$)**

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2.2. Find pseudo-inverse of $\mathbf{A}$. (No need to compute the actual value.)

Key: $A^+ = V\Sigma^+U^T$

$$\therefore A^+ = \begin{bmatrix} 1/\sqrt{3} & 0 & -2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 1/\sqrt{30} & 0 \\ 0 & 1/2\sqrt{5} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix}^T.$$

2.3. Find all eigenvalues of $\mathbf{A^T A}$.

Key: $A^TA = V\Sigma^T\Sigma V^T$

$$\therefore \Sigma^T\Sigma = \begin{bmatrix} \sqrt{30} & 0 \\ 0 & 2\sqrt{5} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{30} & 0 & 0 \\ 0 & 2\sqrt{5} & 0 \end{bmatrix} = \begin{bmatrix} 30 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore A^TA = V \begin{bmatrix} 30 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 0 \end{bmatrix} V^T$$ ⇝ diagonalized form.

$\therefore$ All eigenvalues consist of 30, 20 and 0.

3. (1 point) Do you have any suggestions or comments about the TA team or this subject in general? Feel free to write or draw anything. (Just drawing one single dot is enough for a full credit.)

Don't forget to practice more on final preparation,

wish you guys the best of luck krub. :).